

Healthcare Fraud Provider Analysis

Business Question

What characteristics distinguish fraudulent healthcare providers from non-fraudulent healthcare providers?

Overview

This project analyzes Medicare-style healthcare claims data to identify billing and utilization patterns associated with providers fraud risk. This analysis combines SQL for data cleaning and querying, excel for validation, and Power BI for the final interactive dashboard.

Dataset

- Source: [HEALTHCARE PROVIDER FRAUD DETECTION ANALYSIS](#)(Kaggle CMS-style Medicare Claims)
- 138556- beneficiary, 40474- inpatient, 517737- outpatient, 5410- providers
- Four linked table: Provider, Beneficiary, Inpatient, outpatient claims

Tool Used

- SQL: Schema design, Data cleaning, and analysis queries
- Excel (Power Query, Pivot Table): Data validation and exploration
- Power Bi: Final Interactive Design

Methodology

- Schema Design and Import: Design MySQL schema with foreign key constraints; loaded the data via LOAD DATA INFILE
- Data Cleaning: Identified 27 negative reimbursement amount in beneficiary table (flagged as legitimate claims adjustments rather than removed)

- Analysis: Wrote SQL Queries to test 5 hypotheses using provider level average throughout for fair comparison between fraud and non-fraud providers
- Validation: Cross-checked SQL results from excel PivotTable built from same analytical dataset
- Visualization: Built Power BI Dashboard summarizing findings for a non-technical audience

Key Findings

Questions	Non-Fraud	Fraud	Differences
Avg. claim amount per provider	1523.78	3842.80	2.5x higher
Avg. inpatient claims per provider	10.33	53.19	5x higher
Avg. total claims per provider	5.44	420.55	6x higher
Avg. patient age	70.44	72.93	No meaningful differences
Avg. chronic conditions per patient	5.44	5.39	No meaningful difference

Conclusion

Fraud risk is strongly associated with billing behavior- unusually high total claims volume and claim amount per provider - rather than patients characteristics. Patients age and their chronic condition show no meaningful difference between fraud and non-fraud providers group.

Dashboard

